RentLens: Australia Rent Affordability Analysis $Privatemarket$

I. Introduction

1. Overview

Housing affordability is a major concern across Australia, particularly in capital cities where rent often outpaces household income. RentLens is a Python project that analyzes rental stress $privatemarket$ across all Local Government Areas $LGAs$ using 2021 ABS census data and geospatial information. The project provides insights into how rent-to-income ratios vary with population, housing stock, population density, and distance from capital cities.

2. Objectives

• Calculate weighted average weekly rent, weekly income per LGA
• Classify LGAs by rental affordability $rent_to_incomeratio$: Affordable, Moderately Unaffordable, Severely Unaffordable.
• Explore correlations between rent-to-income ratio, population, housing stock, and density.
• Analyze how proximity to capital cities affects rental stress.
• Visualize patterns using maps, heatmaps and scatter plot.

3. Mothodoligy

• Data processing and analysis with pandas and geopandas.
• Choropleth maps showing affordability categories using matplotlib.
• Correlation heatmaps for population, density, housing stock, and rent stress with seaborn.
• Computation of distance from each LGA to the nearest capital city to study spatial trends.

4. Key Visual Outputs

• Choropleth maps of rental affordability across Australia.
• Heatmaps of correlations between demographic and rental metrics.
• Scatter plots showing rent-to-income ratios relative to distance from capital cities.
• Bivariate and gradient maps highlighting areas of highest rental stress.

5. Impact

RentLens provides a data-driven perspective on housing affordability, helping researchers, policymakers, and the public understand which regions are most under rental stress and how geographic factors influence housing pressures.

II. Data exploration

1. Import inportant libraries

First step is to import some basic libraries used data processing and visualization:**

python

```
!pip install geopandas
# Importing libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import geopandas as gpd
import seaborn as sns
import matplotlib.patches as mpatches
```

```
Requirement already satisfied: geopandas in /local_disk0/.ephemeral_nfs/envs/pythonEnv-e3faff42-e04e-46e9-ac87-
bc140e2383b1/lib/python3.12/site-packages (1.1.2)
Requirement already satisfied: numpy>=1.24 in /databricks/python3/lib/python3.12/site-packages (from geopandas)
(2.1.3)
Requirement already satisfied: pyogrio>=0.7.2 in /local_disk0/.ephemeral_nfs/envs/pythonEnv-e3faff42-e04e-46e9-
ac87-bc140e2383b1/lib/python3.12/site-packages (from geopandas) (0.12.1)
Requirement already satisfied: packaging in /databricks/python3/lib/python3.12/site-packages (from geopandas)
(24.1)
Requirement already satisfied: pandas>=2.0.0 in /databricks/python3/lib/python3.12/site-packages (from geopandas)
(2.2.3)
Requirement already satisfied: pyproj>=3.5.0 in /local_disk0/.ephemeral_nfs/envs/pythonEnv-e3faff42-e04e-46e9-
ac87-bc140e2383b1/lib/python3.12/site-packages (from geopandas) (3.7.2)
Requirement already satisfied: shapely>=2.0.0 in /local_disk0/.ephemeral_nfs/envs/pythonEnv-e3faff42-e04e-46e9-
ac87-bc140e2383b1/lib/python3.12/site-packages (from geopandas) (2.1.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /databricks/python3/lib/python3.12/site-packages (from
pandas>=2.0.0->geopandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /databricks/python3/lib/python3.12/site-packages (from
pandas>=2.0.0->geopandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in /databricks/python3/lib/python3.12/site-packages (from
pandas>=2.0.0->geopandas) (2024.1)
Requirement already satisfied: certifi in /databricks/python3/lib/python3.12/site-packages (from pyogrio>=0.7.2-
>geopandas) (2025.1.31)
Requirement already satisfied: six>=1.5 in /usr/lib/python3/dist-packages (from python-dateutil>=2.8.2-
>pandas>=2.0.0->geopandas) (1.16.0)
[43mNote: you may need to restart the kernel using %restart_python or dbutils.library.restartPython() to use
updated packages.[0m
```

2. Data loading

There are 4 datasets used
- Rent dataset
- Income dataset
- Population dataset
- LGAs shapefile

The first 3 datasets are latest census in 2021, downloaded from
2021 Australia, Census All persons QuickStats | Australian Bureau of Statistics
LGA shapefile is downloaded from Digital boundary file from ABS website $2021 version$
Digital boundary files | Australian Bureau of Statistics.

Rent and income are the most two most important datasets for the project, which are used to calculate key rent_to_income ratio metric.
Next part will discussed about these two dataset in details

a. Rent data

python

```
# Load rent dataset
rent_df = pd.read_csv('/Volumes/workspace/project-trial/rental_private_market/Weekly Rent.csv')
```

python

```
rent_df.shape
```

```
(566, 119)
```

python

```
rent_df.head()
```

	OBJECTID	Local Government Areas 2021 code	Local Government Areas 2021 name	Area in square kilometres	Uniform Resource Identifier	Rent: 1—74: Landlord type: Real estate agent	Rent: 1—74: Landlord type: State or territory housing authority	Rent: 1—74: Landlord type: Community housing provider	Rent: 1—74: Landlord type: Person not in same household	Rent: 1—74: Landlord type: Other landlord type	Rent: 1 7 Landlor typ No state
0	1	10050	Albury	305.6386	https://linked.data.gov.au/dataset/asgsed3/LGA...	13.0	15.0	5.0	5.0	0.0	0
1	2	10180	Armidale Regional	7809.4406	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	3.0	5.0	9.0	10.0	0
2	3	10250	Ballina	484.9692	https://linked.data.gov.au/dataset/asgsed3/LGA...	5.0	9.0	3.0	0.0	5.0	0
3	4	10300	Balranald	21690.7493	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	6.0	3.0	0.0	3.0	0
4	5	10470	Bathurst Regional	3817.8645	https://linked.data.gov.au/dataset/asgsed3/LGA...	4.0	19.0	12.0	5.0	0.0	0

5 rows × 119 columns

The rent dataset has 566 rows and 119 columns, providing counts of rented dwellings grouped into weekly rent ranges $rentbrackets$, each row is for one LGA.

Each rent bracket has 7 columns, six first columns for 6 different landlord types, the 7th column is total of the 6 previous columns

Landlord types:

• Real estate agent: houses rented privately through real agent $hereaftercall\\"privatemarket$
• State or territory housing authority and Community housing provider: rented and managed by government
• Person not in same household: is a person who is enumerated at a private dwelling but is not part of the primary household that occupies that dwelling. This could include lodgers, boarders, or guests who are temporarily or semi-permanently residing at the address.
• Other landlord type: This is a residual category for renters whose landlord does not fit into any of the other, more specific categories. It is a residual category for those who rent but do not

Private rental market vs. social/public housing

When we calculate rental stress using Census data $e.g.averagerent, averageincomebyLGA$, we're mostly measuring the private rental market because:

● The majority of renting households in Australia rent privately, not from government or community housing.

● Social housing tenants $publicorcommunityhousing$ typically pay below-market rents, often capped at around 25% of household income, so they usually don't appear "stressed" under this metric.

This project scope is focused on private rental market $Landlordtype : Realestateagent$

As per 2021 census statistic, in Australia 30.6% household rent, 35% household owned with mortgage and 31% owned outright $thesegroupmayhavemultipleotherinvestmentproperties$

In this project, only renting household income data (**rented group**) is used in order to draw a real picture of renting affordability. This is a sub-set of G33 data set which list income of all types of household.

Income data

python

```
income_df = pd.read_csv("/Volumes/workspace/project-trial/rental_private_market/Income Renting Households.cs
```

python

```
income_df
```

	Census year	Local Government Areas 2021 code	Local Government Areas 2021 name	Renting households	Nil income	$1-149$ $\$1-\$7,799$	$150-299$ $\$7,800-\$15,599$	$300-399$ $\$15,600-\$20,799$	$400-499$ $\$20,800-\$25,999$	$500-649$ $\$26,000-\$33,799$	$650-799$ $\$33,800-\$41,599$	$800-999$ $\$41,600-\$51,999$	$1,000-1,249$ $\$52,000-\$64,999$	$1,250-1,499$ $\$65,000-\$77,999$	$1,500-1,749$ $\$78,000-\$90,999$	$1,750-1,999$ $\$91,000-\$103,999$	$2,000$ or more $\$104,000$ or more	Hours worked not stated	Total
0	LGA 2021	10050	Albury	Total renting households	116	61	292	553	848	1040	1000	1000	1000	1000	1000	1000	1000	1000	1000
1	LGA 2021	10180	Armidale Regional	Total renting households	60	40	196	282	358	520	520	520	520	520	520	520	520	520	520
2	LGA 2021	10250	Ballina	Total renting households	61	46	122	225	498	654	654	654	654	654	654	654	654	654	654
3	LGA 2021	10300	Balranald	Total renting households	4	0	9	11	18	21	21	21	21	21	21	21	21	21	21
4	LGA 2021	10470	Bathurst Regional	Total renting households	56	29	161	284	468	595	595	595	595	595	595	595	595	595	595
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
551	LGA 2021	79799	Migratory - Offshore - Shipping NT	Total renting households	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
552	LGA 2021	89399	Unincorporated ACT	Total renting households	643	301	1311	2036	3141	2915	2915	2915	2915	2915	2915	2915	2915	2915	2915
553	LGA 2021	89799	Migratory - Offshore - Shipping ACT	Total renting households	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
554	LGA 2021	99399	Unincorp. Other Territories	Total renting households	4	0	13	13	33	42	42	42	42	42	42	42	42	42	42
555	LGA 2021	99799	Migratory - Offshore - Shipping OT	Total renting households	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

556 rows × 24 columns

The income dataset has 556 rows and 24 columns, providing counts of dwellings grouped into weekly income ranges $income brackets\ weekly amount and equivalent annual income$, each row is for one LGA.

Basic information and Checking null values

python

```
# Basic information
rent_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 566 entries, 0 to 565
Columns: 119 entries, OBJECTID to Shape__Length
dtypes: float64(114), int64(1), object(4)
memory usage: 526.3+ KB
```

python

```
rent_df.dtypes
```

```
OBJECTID                                  int64
Local Government Areas 2021 code         object
Local Government Areas 2021 name         object
Area in square kilometres               float64
Uniform Resource Identifier              object
                                          ...
Rent total: Landlord type: Not stated   float64
Rent total: Total                       float64
globalid                                 object
Shape__Area                             float64
Shape__Length                           float64
Length: 119, dtype: object
```

python

```
rent_df.shape
```

```
(566, 119)
```

python

```
income_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 556 entries, 0 to 555
Data columns (total 24 columns):
 #   Column                              Non-Null Count  Dtype 
---  ------                              --------------  ----- 
 0   Census year                         556 non-null    object
 1   Local Government Areas 2021 code    556 non-null    int64 
 2   Local Government Areas 2021 name    556 non-null    object
 3   Renting households                  556 non-null    object
 4   Nil income                          556 non-null    int64 
 5   $1-$149 ($1-$7,799)                 556 non-null    int64 
 6   $150-$299 ($7,800-$15,599)          556 non-null    int64 
 7   $300-$399 ($15,600-$20,799)         556 non-null    int64 
 8   $400-$499 ($20,800-$25,999)         556 non-null    int64 
 9   $500-$649 ($26,000-$33,799)         556 non-null    int64 
 10  $650-$799 ($33,800-$41,599)         556 non-null    int64 
 11  $800-$999 ($41,600-$51,999)         556 non-null    int64 
 12  $1,000-$1,249 ($52,000-$64,999)     556 non-null    int64 
 13  $1,250-$1,499 ($65,000-$77,999)     556 non-null    int64 
 14  $1,500-$1,749 ($78,000-$90,999)     556 non-null    int64 
 15  $1,750-$1,999 ($91,000-$103,999)    556 non-null    int64 
 16  $2,000-$2,499 ($104,000-$129,999)   556 non-null    int64 
 17  $2,500-$2,999 ($130,000-$155,999)   556 non-null    int64 
 18  $3,000-$3,499 ($156,000-$181,999)   556 non-null    int64 
 19  $3,500 or more ($182,000 or more)   556 non-null    int64 
 20  Partial income stated               556 non-null    int64 
 21  All incomes not stated              556 non-null    int64 
 22  Not applicable                      556 non-null    int64 
 23  Total                               556 non-null    int64 
dtypes: int64(21), object(3)
memory usage: 104.4+ KB
```

python

```
rent_df.isnull().sum().sort_values(ascending=False)
```

```
Shape__Length                                        19
Shape__Area                                          19
Area in square kilometres                            19
Rent: $650-$749: Landlord type: Real estate agent     1
Rent: $750-$849: Landlord type: Not stated            1
                                                     ..
globalid                                              0
Local Government Areas 2021 name                      0
Local Government Areas 2021 code                      0
Uniform Resource Identifier                           0
OBJECTID                                              0
Length: 119, dtype: int64
```

python

```
missing_lga = rent_df[rent_df["Rent: $650-$749: Landlord type: Real estate agent"].isnull()]

missing_lga[["Local Government Areas 2021 code",
             "Local Government Areas 2021 name"]]
```

	Local Government Areas 2021 code	Local Government Areas 2021 name
565	ZZZZZ	Outside Australia

There are 5 missing fields with missing data. As this is rent dataframe, the only concerning field is Rent: $650-749$ and Rent: $750-849$: Landlord type: Real estate agent" which belongs to Local Government - ZZZZZ Outside Australia. This is used when a Census table needs to include people, dwellings, or data that are not assigned to any Australian geographic area because the person's usual residence is overs.

python

```
# Checking null value of income df

income_df.isnull().sum().value_counts()
```

```
0    24
Name: count, dtype: int64
```

There's no null value in income income data

Identify 10 rows $LGAs$ difference between rent_df and icome_df

Rent data has 566 rows and income data has 556 rows. To identify the row difference, left join is used with rent data as left table.

LGA code is used as keyword to merge, before merging we convert LGAs codes from 2 datasets to same datatype $string$

python

```
## rent has 566 rows, income has 556 rows - 10 rows difference
rent_df['Local Government Areas 2021 code'] = rent_df['Local Government Areas 2021 code'].astype(str)
income_df['Local Government Areas 2021 code'] = income_df['Local Government Areas 2021 code'].astype(str)

check_len = rent_df.merge(income_df, how = 'left', on = 'Local Government Areas 2021 code', indicator = True
```

python

```
check_len
```

	OBJECTID	Local Government Areas 2021 code	Local Government Areas 2021 name_x	Area in square kilometres	Uniform Resource Identifier	Rent: 1–74: Landlord type: Real estate agent	Rent: 1–74: Landlord type: State or territory housing authority	Rent: 1–74: Landlord type: Community housing provider	Rent: 1–74: Landlord type: Person not in same household	Rent: 1–74: Landlord type: Other landlord type	Rent: Land t sta
0	1	10050	Albury	305.6386	https://linked.data.gov.au/dataset/asgsed3/LGA...	13.0	15.0	5.0	5.0	0.0	
1	2	10180	Armidale Regional	7809.4406	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	3.0	5.0	9.0	10.0	
2	3	10250	Ballina	484.9692	https://linked.data.gov.au/dataset/asgsed3/LGA...	5.0	9.0	3.0	0.0	5.0	
3	4	10300	Balranald	21690.7493	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	6.0	3.0	0.0	3.0	
4	5	10470	Bathurst Regional	3817.8645	https://linked.data.gov.au/dataset/asgsed3/LGA...	4.0	19.0	12.0	5.0	0.0	
...	...	...	...	...	...	...	...	...	...	...	
561	562	89799	Migratory - Offshore - Shipping ACT	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
562	563	99399	Unincorp. Other Territories	105.8806	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	23.0	5.0	4.0	
563	564	99499	No usual address OT	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
564	565	99799	Migratory - Offshore - Shipping OT	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
565	566	ZZZZZ	Outside Australia	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	NaN	NaN	NaN	NaN	NaN	

566 rows × 143 columns

python

```
check_len._merge.value_counts()
```

```
_merge
both          556
left_only      10
right_only      0
Name: count, dtype: int64
```

python

```
check_len[check_len['_merge'] =='left_only']
```

	OBJECTID	Local Government Areas 2021 code	Local Government Areas 2021 name_x	Area in square kilometres	Uniform Resource Identifier	Rent: 1–74: Landlord type: Real estate agent	Rent: 1–74: Landlord type: State or territory housing authority	Rent: 1–74: Landlord type: Community housing provider	Rent: 1–74: Landlord type: Person not in same household	Rent: 1–74: Landlord type: Other landlord type	Rent: Landl ty sta
129	130	19499	No usual address NSW	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
211	212	29499	No usual address $Vic.$	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
291	292	39499	No usual address Qld	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
364	365	49499	No usual address SA	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
505	506	59499	No usual address WA	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
536	537	69499	No usual address $Tas.$	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
557	558	79499	No usual address NT	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
560	561	89499	No usual address ACT	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
563	564	99499	No usual address OT	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0	
565	566	ZZZZZ	Outside Australia	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	NaN	NaN	NaN	NaN	NaN	

10 rows × 143 columns

The 10 LGAs are in rent_df and not in income_df is as above.

Rent data of ZZZZ LGA – Outside Australia is null as mentioned in previous section
Other 9 rows are not actual LGAs, it's an ABS identifiers for "No usual address" population, who may be homeless, rough sleepers. These 10 rows are not important to this analysis.

Removing all other landlord type out of rent_data, only keep "Real estate agent" data $privatemarket$
A "for loop" is used is used to show all columns containing the phrase "landlord type: real estate agent" $case-insensitive$

python

```
[c for c in rent_df.columns if "landlord type: real estate agent" in c.lower()]
```

```
['Rent: $1–$74: Landlord type: Real estate agent',
 'Rent: $75–$99: Landlord type: Real estate agent',
 'Rent: $100–$149: Landlord type: Real estate agent',
 'Rent: $150–$199: Landlord type: Real estate agent',
 'Rent: $200–$224: Landlord type: Real estate agent',
 'Rent: $225–$274: Landlord type: Real estate agent',
 'Rent: $275–$349: Landlord type: Real estate agent',
 'Rent: $350–$449: Landlord type: Real estate agent',
 'Rent: $450–$549: Landlord type: Real estate agent',
 'Rent: $550–$649: Landlord type: Real estate agent',
 'Rent: $650–$749: Landlord type: Real estate agent',
 'Rent: $750–$849: Landlord type: Real estate agent',
 'Rent: $850–$949: Landlord type: Real estate agent',
 'Rent: $950 and over: Landlord type: Real estate agent',
 'Rent: Rent not stated: Landlord type: Real estate agent',
 'Rent total: Landlord type: Real estate agent']
```

Then, a key-value pairs dictionary is created, with keys are from previous step $cloop$. This to ensure 100% keyword accuracy). Value is mid-point of rent bracket $whichismanuallyentered.$ The reason midpoint is added, it is because ABS rent brackets are ranges, to calculate average weekly rent in later steps, and we need a numeric value, not a text label.

python

```
## Creating a dictionary
rent_brackets = {
    "Rent: $1-$74: Landlord type: Real estate agent": 37.5,
    "Rent: $75-$99: Landlord type: Real estate agent": 87,
    "Rent: $100-$149: Landlord type: Real estate agent": 125,
    "Rent: $150-$199: Landlord type: Real estate agent": 175,
    "Rent: $200-$224: Landlord type: Real estate agent": 212,
    "Rent: $225-$274: Landlord type: Real estate agent": 250,
    "Rent: $275-$349: Landlord type: Real estate agent": 312,
    "Rent: $350-$449: Landlord type: Real estate agent": 400,
    "Rent: $450-$549: Landlord type: Real estate agent": 500,
    "Rent: $550-$649: Landlord type: Real estate agent": 600,
    "Rent: $650-$749: Landlord type: Real estate agent": 700,
    "Rent: $750-$849: Landlord type: Real estate agent": 800,
    "Rent: $850-$949: Landlord type: Real estate agent": 900,
    "Rent: $950 and over: Landlord type: Real estate agent": 1025,

}

rent_cols = [c for c in rent_df.columns if c in rent_brackets]
```

python

```
rent_cols
```

```
['Rent: $1-$74: Landlord type: Real estate agent',
 'Rent: $75-$99: Landlord type: Real estate agent',
 'Rent: $100-$149: Landlord type: Real estate agent',
 'Rent: $150-$199: Landlord type: Real estate agent',
 'Rent: $200-$224: Landlord type: Real estate agent',
 'Rent: $225-$274: Landlord type: Real estate agent',
 'Rent: $275-$349: Landlord type: Real estate agent',
 'Rent: $350-$449: Landlord type: Real estate agent',
 'Rent: $450-$549: Landlord type: Real estate agent',
 'Rent: $550-$649: Landlord type: Real estate agent',
 'Rent: $650-$749: Landlord type: Real estate agent',
 'Rent: $750-$849: Landlord type: Real estate agent',
 'Rent: $850-$949: Landlord type: Real estate agent',
 'Rent: $950 and over: Landlord type: Real estate agent']
```

python

```
freq_rent = rent_df[rent_cols]
```

python

```
freq_rent
```

	Rent: $1–74: Landlord type: Real estate agent	Rent: $75–99: Landlord type: Real estate agent	Rent: $100–149: Landlord type: Real estate agent	Rent: $150–199: Landlord type: Real estate agent	Rent: $200–224: Landlord type: Real estate agent	Rent: $225–274: Landlord type: Real estate agent	Rent: $275–349: Landlord type: Real estate agent	Rent: $350–449: Landlord type: Real estate agent	Rent: $450–549: Landlord type: Real estate agent	Rent: $550–649: Landlord type: Real estate agent	Rent: $650–749: Landlord type: Real estate agent	Rent: $750–849: Landlord type: Real estate agent	Rent: $850–949: Landlord type: Real estate agent	Rent: $950 and over: Landlord type: Real estate agent
0	13.0	10.0	100.0	759.0	480.0	967.0	1463.0	1149.0	274.0	60.0	16.0	4.0	0.0	5.0
1	0.0	6.0	35.0	139.0	205.0	459.0	688.0	688.0	155.0	31.0	8.0	0.0	0.0	3.0
2	5.0	0.0	9.0	14.0	11.0	60.0	292.0	735.0	685.0	527.0	305.0	78.0	37.0	23.0
3	0.0	0.0	0.0	14.0	7.0	6.0	9.0	4.0	0.0	0.0	0.0	0.0	0.0	0.0
4	4.0	0.0	32.0	138.0	139.0	374.0	957.0	1158.0	409.0	37.0	3.0	6.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
561	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
562	0.0	0.0	0.0	15.0	8.0	13.0	35.0	22.0	6.0	0.0	0.0	0.0	0.0	0.0
563	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
564	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
565	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

566 rows × 14 columns

Rent_cols now is a list consisting of Real estate agent data only.

Plotting to show which rent brackets have the most dwellings.

python

```
# Compute total dwellings per bracket across all LGAs
bracket_totals = rent_df[rent_cols].sum()

# Plot
plt.figure(figsize=(12,6))
plt.bar(bracket_totals.index, bracket_totals.values)
plt.xticks(rotation=90)
plt.xlabel("Rent brackets")
plt.ylabel("Total dwellings")
plt.title("Distribution of dwellings by rent bracket (Real estate agent landlord)")
plt.tight_layout()
plt.show()
```

Rent bracket $\$349 - \$399$ has the highest number of dwellings $600,000 houses$, followed by $\$450 - \$549$ bracket with around 360,000 houses and $\$275 - \$349$ bracket with total of houses at 310 000

Average weeky rent

To calculate average weekly rent from the ABS 2021 Census rent brackets, we must use a midpoint weighted average — because ABS provides counts of households in rent brackets, not exact rents.

Avg weekly rent = Σ $midpoint of bracket \times households$ / Σ $households$

python

```
midpoints = [rent_brackets[c] for c in rent_cols]
```

python

```
midpoints
```

```
[37.5, 87, 125, 175, 212, 250, 312, 400, 500, 600, 700, 800, 900, 1025]
```

python

```
rent_df["avg_rent_weekly"] = (freq_rent * midpoints).sum(axis=1) / freq_rent.sum(axis=1)
```

python

```
rent_df = rent_df[["Local Government Areas 2021 code", "Local Government Areas 2021 name", "avg_rent_weekly"
rent_df
```

	Local Government Areas 2021 code	Local Government Areas 2021 name	avg_rent_weekly
0	10050	Albury	301.655377
1	10180	Armidale Regional	323.567646
2	10250	Ballina	502.579468
3	10300	Balranald	246.050000
4	10470	Bathurst Regional	352.057722
...	...	...	...
561	89799	Migratory - Offshore - Shipping ACT	NaN
562	99399	Unincorp. Other Territories	305.969697
563	99499	No usual address OT	NaN
564	99799	Migratory - Offshore - Shipping OT	NaN
565	ZZZZZ	Outside Australia	NaN

566 rows × 3 columns

Average weekly `data`

Caculation is done similar to average weekly rent

python

```python
brackets = {
    'Nil income': 0,
    '$1-$149 ($1-$7,799)': 75,
    '$150-$299 ($7,800-$15,599)': 225,
    '$300-$399 ($15,600-$20,799)': 350,
    '$400-$499 ($20,800-$25,999)': 450,
    '$500-$649 ($26,000-$33,799)': 575,
    '$650-$799 ($33,800-$41,599)': 725,
    '$800-$999 ($41,600-$51,999)': 900,
    '$1,000-$1,249 ($52,000-$64,999)': 1125,
    '$1,250-$1,499 ($65,000-$77,999)': 1350,
    '$1,500-$1,749 ($78,000-$90,999)': 1625,
    '$1,750-$1,999 ($91,000-$103,999)': 1875,
    '$2,000-$2,499 ($104,000-$129,999)': 2250,
    '$2,500-$2,999 ($130,000-$155,999)': 2750,
    '$3,000-$3,499 ($156,000-$181,999)': 3250,
    '$3,500 or more ($182,000 or more)': 4000,

}

# Create frq dataframe of only income related columns ~ 16 columns at differnt bracket (there are 24 columns
income_cols = [c for c in income_df.columns if c in brackets]
freq = income_df[income_cols]

# Average income per LGA
income_df['avg_income_weekly'] = (freq * [brackets[c] for c in income_cols]).sum(axis=1) / freq.sum(axis=1)

income_df[['Local Government Areas 2021 code', 'Local Government Areas 2021 name', 'avg_income_weekly']]
```

	Local Government Areas 2021 code	Local Government Areas 2021 name	avg_income_weekly
0	10050	Albury	917.772193
1	10180	Armidale Regional	930.153891
2	10250	Ballina	1013.969938
3	10300	Balranald	1007.601351
4	10470	Bathurst Regional	990.844444
...	...	...	...
551	79799	Migratory - Offshore - Shipping NT	NaN
552	89399	Unincorporated ACT	1410.514539
553	89799	Migratory - Offshore - Shipping ACT	NaN
554	99399	Unincorp. Other Territories	1003.785211
555	99799	Migratory - Offshore - Shipping OT	NaN

556 rows × 3 columns

python

```
totals = freq.sum()

plt.figure(figsize=(12,12))
plt.bar(totals.index, totals.values)
plt.xticks(rotation=90)
plt.xlabel("Income bracket")
plt.ylabel("Number of households")
plt.title("Total renting households by income bracket")
plt.tight_layout()
plt.show()
```

The top 3 popular weekly income is $1000—1249$ with around 33,000 households, followed by $800-999 bracket into which 29,000 household fall into and 500-$649 bracket to which 27,000 dwellings belong to.

III. Rent_to_income ratio

Rent_to_income ratio is the most important metric in this project, which will define rental affordability.

Average weekly rent and income are calculated in section II. Rent_df and income_df have additional “avg_weekly_rent” and “avg_weekly_income” as last column.
To calculate rent_to_income ratio, we merge 2 dataframes using INNER JOIN as we already know that income data has 556 rows and rent data has 566 rows. Ten rows difference are LGAs created for people living overseas or people with no usual address. There 10 groups are not important to the project.
LGA code are converted to same datatype $string$ before merging.

python

```
rent_df['Local Government Areas 2021 code'] = rent_df['Local Government Areas 2021 code'].astype(str)
income_df['Local Government Areas 2021 code'] = income_df['Local Government Areas 2021 code'].astype(str)

merged = pd.merge(rent_df, income_df, on="Local Government Areas 2021 code", how="inner")

# Calculate rent to income ratio
merged["rent_to_income_ratio"] = merged["avg_rent_weekly"] / merged["avg_income_weekly"]
```

python

```
merged
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	avg_rent_weekly	Census year	Local Government Areas 2021 name_y	Renting households	Nil income	$1-149$ $\$1-\$7,799$	$150-299$ $\$7,800-\$15,599$	$300-399$ $\$15,600-\$20,799$	4 $\$20,800-$
0	10050	Albury	301.655377	LGA 2021	Albury	Total renting households	116	61	292	553	
1	10180	Armidale Regional	323.567646	LGA 2021	Armidale Regional	Total renting households	60	40	196	282	
2	10250	Ballina	502.579468	LGA 2021	Ballina	Total renting households	61	46	122	225	
3	10300	Balranald	246.050000	LGA 2021	Balranald	Total renting households	4	0	9	11	
4	10470	Bathurst Regional	352.057722	LGA 2021	Bathurst Regional	Total renting households	56	29	161	284	
...	...	...	...	...	...	...	...	...	...	...	
551	79799	Migratory - Offshore - Shipping NT	NaN	LGA 2021	Migratory - Offshore - Shipping NT	Total renting households	0	0	0	0	
552	89399	Unincorporated ACT	529.252381	LGA 2021	Unincorporated ACT	Total renting households	643	301	1311	2036	
553	89799	Migratory - Offshore - Shipping ACT	NaN	LGA 2021	Migratory - Offshore - Shipping ACT	Total renting households	0	0	0	0	
554	99399	Unincorp. Other Territories	305.969697	LGA 2021	Unincorp. Other Territories	Total renting households	4	0	13	13	
555	99799	Migratory - Offshore - Shipping OT	NaN	LGA 2021	Migratory - Offshore - Shipping OT	Total renting households	0	0	0	0	

556 rows × 28 columns

Merged data now has 556 rows, represent 556 LGAs and the 28th column is rent_to_income ratio

IV. Rental affordability classification

1. Threshold 30% and 38%

Threshold 30% is widely used in mortgage affordably by Banks and rent affordably by Real estate Company

In recent years, 38% threshold has been used as “severely unfordable”. Studies from the US, Canada, Australia, and the UK show m ortgage/rent burdens above 38% strongly predict payment arrears or bankruptcy.

2. We categorize affordability into 3 level using rent_to_income_ratio:

- Rent_to_income ratio <= 30%, Affordable
- Rent_to_income ratio > 30% and <= 38%: Moderately unaffordable
- Rent_to_income ratio > 38%: Severely unaffordable

python

```
merged["affordability"] = np.where(merged["rent_to_income_ratio"] >0.38, "Severely Unaffordable", np.where(m
```

python

```
merged
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	avg_rent_weekly	Census year	Local Government Areas 2021 name_y	Renting households	Nil income	1−149 $1 − $7,799	150−299 $7,800 − $15,599	300−399 $15,600 − $20,799	4 $20,800 −
0	10050	Albury	301.655377	LGA 2021	Albury	Total renting households	116	61	292	553	
1	10180	Armidale Regional	323.567646	LGA 2021	Armidale Regional	Total renting households	60	40	196	282	
2	10250	Ballina	502.579468	LGA 2021	Ballina	Total renting households	61	46	122	225	
3	10300	Balranald	246.050000	LGA 2021	Balranald	Total renting households	4	0	9	11	
4	10470	Bathurst Regional	352.057722	LGA 2021	Bathurst Regional	Total renting households	56	29	161	284	
...	...	...	...	...	...	...	...	...	...	...	
551	79799	Migratory - Offshore - Shipping NT	NaN	LGA 2021	Migratory - Offshore - Shipping NT	Total renting households	0	0	0	0	
552	89399	Unincorporated ACT	529.252381	LGA 2021	Unincorporated ACT	Total renting households	643	301	1311	2036	
553	89799	Migratory - Offshore - Shipping ACT	NaN	LGA 2021	Migratory - Offshore - Shipping ACT	Total renting households	0	0	0	0	
554	99399	Unincorp. Other Territories	305.969697	LGA 2021	Unincorp. Other Territories	Total renting households	4	0	13	13	
555	99799	Migratory - Offshore - Shipping OT	NaN	LGA 2021	Migratory - Offshore - Shipping OT	Total renting households	0	0	0	0	

556 rows × 29 columns

python

```
merged[['Local Government Areas 2021 code','Local Government Areas 2021 name_x','avg_rent_weekly', 'avg_inco
       'rent_to_income_ratio', 'affordability']]
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	avg_rent_weekly	avg_income_weekly	rent_to_income_ratio	affordability
0	10050	Albury	301.655377	917.772193	0.328682	Moderately Unaffordable
1	10180	Armidale Regional	323.567646	930.153891	0.347865	Moderately Unaffordable
2	10250	Ballina	502.579468	1013.969938	0.495655	Severely Unaffordable
3	10300	Balranald	246.050000	1007.601351	0.244194	Affordable
4	10470	Bathurst Regional	352.057722	990.844444	0.355311	Moderately Unaffordable
...	...	...	...	...	...	...
551	79799	Migratory - Offshore - Shipping NT	NaN	NaN	NaN	No data
552	89399	Unincorporated ACT	529.252381	1410.514539	0.375219	Moderately Unaffordable
553	89799	Migratory - Offshore - Shipping ACT	NaN	NaN	NaN	No data
554	99399	Unincorp. Other Territories	305.969697	1003.785211	0.304816	Moderately Unaffordable
555	99799	Migratory - Offshore - Shipping OT	NaN	NaN	NaN	No data

556 rows × 6 columns

python

```
merged.affordability.value_counts()
```

```
affordability
Moderately Unaffordable    185
Severely Unaffordable      156
Affordable                 122
No data                     93
Name: count, dtype: int64
```

python

```
labels = ['Moderately Unaffordable', 'Severely unaffordable', 'Affordable', 'No data']
sizes = [185, 156, 122, 93]

# Plot pie chart
plt.figure(figsize=(7,7))
plt.pie(sizes, labels=labels, autopct='%1.1f%%')  # no manual colors specified
plt.title("Rental Affordability Across Australia LGAs")
plt.show()
```

Among 556 LGAs in merged dataframe, there are

- 156 LGAs are classified as “severely unaffordable”, account for 28.1%.
- 185 LGAs are classified as “moderately affordable”, account for 33.3%
- 122 LGAs are categorized as “affordable”, account for 21.9%
- 93 LGAs with “No data”

python

```
merged[merged['affordability'] =='No data']
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	avg_rent_weekly	Census year	Local Government Areas 2021 name_y	Renting households	Nil income	$1-149$ $\$1-\$7,799$	$150-299$ $\$7,800-\$15,599$	$300-399$ $\$15,600-\$20,799$	4 $\$20,800-$
128	19399	Unincorporated NSW	NaN	LGA 2021	Unincorporated NSW	Total renting households	0	0	0	4	
129	19799	Migratory - Offshore - Shipping NSW	NaN	LGA 2021	Migratory - Offshore - Shipping NSW	Total renting households	0	0	0	0	
209	29399	Unincorporated Vic	NaN	LGA 2021	Unincorporated Vic	Total renting households	0	0	0	0	
210	29799	Migratory - Offshore - Shipping $Vic.$	NaN	LGA 2021	Migratory - Offshore - Shipping $Vic.$	Total renting households	0	0	0	0	
211	30250	Aurukun	NaN	LGA 2021	Aurukun	Total renting households	0	5	22	41	
...	...	...	...	...	...	...	...	...	...	...	
548	74660	West Arnhem	NaN	LGA 2021	West Arnhem	Total renting households	3	28	111	151	
549	74680	West Daly	NaN	LGA 2021	West Daly	Total renting households	0	99	76	45	
551	79799	Migratory - Offshore - Shipping NT	NaN	LGA 2021	Migratory - Offshore - Shipping NT	Total renting households	0	0	0	0	
553	89799	Migratory - Offshore - Shipping ACT	NaN	LGA 2021	Migratory - Offshore - Shipping ACT	Total renting households	0	0	0	0	
555	99799	Migratory - Offshore - Shipping OT	NaN	LGA 2021	Migratory - Offshore - Shipping OT	Total renting households	0	0	0	0	

93 rows × 29 columns

python

```
no_rent_data= merged[merged['affordability'] =='No data'].avg_rent_weekly.isnull().sum()
```

python

```
no_rent_data
```

```
np.int64(93)
```

93 No data LGA are divided into 3 groups:

- First group: Migratory – Offshore – Shipping" refers to people who were counted on census night, on a ship, oil rigs...
- Second group: "Unincorporated" means areas that are not governed by any Local Government Area LGA — so they do not have a council/local municipal government. NSW: Lord Howe, VIC: Alpine Resosrt, Whole ACT is Unoncorporated
- Third group: LGAs in very remote areas which are home of Aboriginals, for example LGA code 74660 – West Arnhem lies right on top of Northern Territory. In such areas, there are no houses privately rented, they're social houses

python

```
merged[['Local Government Areas 2021 code','Local Government Areas 2021 name_x','avg_rent_weekly', 'avg_inco
       'rent_to_income_ratio', 'affordability']]
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	avg_rent_weekly	avg_income_weekly	rent_to_income_ratio	affordability
0	10050	Albury	301.655377	917.772193	0.328682	Moderately Unaffordable
1	10180	Armidale Regional	323.567646	930.153891	0.347865	Moderately Unaffordable
2	10250	Ballina	502.579468	1013.969938	0.495655	Severely Unaffordable
3	10300	Balranald	246.050000	1007.601351	0.244194	Affordable
4	10470	Bathurst Regional	352.057722	990.844444	0.355311	Moderately Unaffordable
...	...	...	...	...	...	...
551	79799	Migratory - Offshore - Shipping NT	NaN	NaN	NaN	No data
552	89399	Unincorporated ACT	529.252381	1410.514539	0.375219	Moderately Unaffordable
553	89799	Migratory - Offshore - Shipping ACT	NaN	NaN	NaN	No data
554	99399	Unincorp. Other Territories	305.969697	1003.785211	0.304816	Moderately Unaffordable
555	99799	Migratory - Offshore - Shipping OT	NaN	NaN	NaN	No data

556 rows × 6 columns

Top stressed LGAs

python

```
top_stressed= (merged[['Local Government Areas 2021 code','Local Government Areas 2021 name_x','avg_rent_wee

)
```

python

```
top_stressed
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	avg_rent_weekly	avg_income_weekly	rent_to_income_ratio
40	12850	Fairfield	440.458050	698.463903	0.630610
18	11350	Byron	653.446709	1125.465149	0.580601
23	11570	Canterbury-Bankstown	478.637966	879.111243	0.544457
110	17550	Tweed	498.469499	928.795547	0.536684
503	60410	Brighton	372.796140	711.343931	0.524073
...	...	...	...	...	...
548	74660	West Arnhem	NaN	846.895641	NaN
549	74680	West Daly	NaN	433.220339	NaN
551	79799	Migratory - Offshore - Shipping NT	NaN	NaN	NaN
553	89799	Migratory - Offshore - Shipping ACT	NaN	NaN	NaN
555	99799	Migratory - Offshore - Shipping OT	NaN	NaN	NaN

556 rows × 5 columns

Fairfield leads as highest rent_to_income ratio. Average weekly income is $698, which means it's quite a low income area however average rent is upto 440/week.
Following Fairfield are Byron, Cantebury- Bankstown and Tweed.

All DA08 class members are living in "moderately unaffordable" and "severely unaffordable" LGAs, led by Cumberland at 0.49 ratio

python

```
da08_lga= merged[merged['Local Government Areas 2021 code'].isin(['12380', '15950', '17200', '11300', '46300
```

python

```
da08_lga=da08_lga[['Local Government Areas 2021 code','Local Government Areas 2021 name_x', 'avg_rent_weekly
```

python

```
da08_lga
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	avg_rent_weekly	avg_income_weekly	rent_to_income_ratio
35	12380	Cumberland	462.245344	950.464441	0.486336
17	11300	Burwood	524.733452	1155.051834	0.454294
290	40070	Adelaide	442.091119	1165.241314	0.379399
105	17200	Sydney	621.990724	1715.249883	0.362624
141	21370	Campaspe	303.897870	850.851900	0.357169
337	46300	Port Lincoln	290.673939	837.549519	0.347053
173	24600	Melbourne	444.877107	1337.058321	0.332728
85	15950	North Sydney	633.736440	2066.515049	0.306669

python

```
merged_map=merged[['Local Government Areas 2021 code','Local Government Areas 2021 name_x','avg_rent_weekly'
       'rent_to_income_ratio', 'affordability']]
```

Show affordability categorization on Australia map

A shorter data frame is created, called “merged_map” which only contain 6 fields, LGA code, name, average weekly rent, average weekly income, rent_to_income ration and affordability classification.

Merged_map has 556 rows, shapefile data has 566 row $10 rows different are for Nous all address and overseas population, which is not important to project.$ "Left join" is still used as NaN LGAs will be grey-color coded.

Color code is used

- Red: severely unaffordable LGAs
- Orange: moderately unaffordable LGA
- Green: affordable LGA
- Light grey: LGA with No data $93 LGAs above mentioned$

python

```
merged_map
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	avg_rent_weekly	avg_income_weekly	rent_to_income_ratio	affordability
0	10050	Albury	301.655377	917.772193	0.328682	Moderately Unaffordable
1	10180	Armidale Regional	323.567646	930.153891	0.347865	Moderately Unaffordable
2	10250	Ballina	502.579468	1013.969938	0.495655	Severely Unaffordable
3	10300	Balranald	246.050000	1007.601351	0.244194	Affordable
4	10470	Bathurst Regional	352.057722	990.844444	0.355311	Moderately Unaffordable
...	...	...	...	...	...	...
551	79799	Migratory - Offshore - Shipping NT	NaN	NaN	NaN	No data
552	89399	Unincorporated ACT	529.252381	1410.514539	0.375219	Moderately Unaffordable
553	89799	Migratory - Offshore - Shipping ACT	NaN	NaN	NaN	No data
554	99399	Unincorp. Other Territories	305.969697	1003.785211	0.304816	Moderately Unaffordable
555	99799	Migratory - Offshore - Shipping OT	NaN	NaN	NaN	No data

556 rows × 6 columns

python

```
lga_gdf = gpd.read_file("/Volumes/workspace/project-trial/rental_private_market/LGA_2021_AUST_GDA2020.shp")
```

python

```
lga_gdf.shape
```

```
(566, 11)
```

python

```
lga_gdf.head()
```

	LGA_CODE21	LGA_NAME21	STE_CODE21	STE_NAME21	AUS_CODE21	AUS_NAME21	AREASQKM21	LOCI_URI21	SHAPE_Leng
0	10050	Albury	1	New South Wales	AUS	Australia	305.6386	http://linked.data.gov.au/dataset/asgsed3/LGA2...	1.321768
1	10180	Armidale Regional	1	New South Wales	AUS	Australia	7809.4406	http://linked.data.gov.au/dataset/asgsed3/LGA2...	6.034583
2	10250	Ballina	1	New South Wales	AUS	Australia	484.9692	http://linked.data.gov.au/dataset/asgsed3/LGA2...	1.511121
3	10300	Balranald	1	New South Wales	AUS	Australia	21690.7493	http://linked.data.gov.au/dataset/asgsed3/LGA2...	11.489912
4	10470	Bathurst Regional	1	New South Wales	AUS	Australia	3817.8645	http://linked.data.gov.au/dataset/asgsed3/LGA2...	5.395114

python

```python
# Load LGA shapefile

lga_gdf = gpd.read_file("/Volumes/workspace/project-trial/rental_private_market/LGA_2021_AUST_GDA2020.shp")

# Load rental/affordability data
merged_map=merged[['Local Government Areas 2021 code','Local Government Areas 2021 name_x','avg_rent_weekly'
       'rent_to_income_ratio', 'affordability']]

#Standardize 5-digit LGA codes

lga_gdf["LGA_CODE21"] = lga_gdf["LGA_CODE21"].astype(str).str.zfill(5)
merged["Local Government Areas 2021 code"] = merged["Local Government Areas 2021 code"].astype(str).str.zfil

# Merge shapefile with rental data
merged_ratio_map = lga_gdf.merge(
    merged_map,
    left_on="LGA_CODE21",
    right_on="Local Government Areas 2021 code",
    how="left"   # keeps all LGAs in shapedata, missing data becomes NaN
)

# Reset index to avoid misalignment
merged_ratio_map = merged_ratio_map.reset_index(drop=True)

#  Handle missing or invalid geometries

merged_ratio_map = merged_ratio_map[merged_ratio_map.geometry.notna()]
merged_ratio_map = merged_ratio_map[merged_ratio_map.geometry.type.isin(['Polygon', 'MultiPolygon'])]
merged_ratio_map = merged_ratio_map.reset_index(drop=True)

# Make 'affordability' categorical

category_order = ["Affordable", "Moderately Unaffordable", "Severely Unaffordable"]
colors = {
    "Affordable": "#2ca02c", # Green
    "Moderately Unaffordable": "#ff7f0e", #Orange
    "Severely Unaffordable": "#d62728" # Red
}

merged_ratio_map['affordability'] = pd.Categorical(
    merged_ratio_map['affordability'],
    categories=category_order,
    ordered=True
)

# Plot map

fig, ax = plt.subplots(1, 1, figsize=(14, 12))

# Plot each affordability category
for cat, color in colors.items():
    subset = merged_ratio_map[merged_ratio_map['affordability'] == cat]
    if len(subset) > 0:
        subset.plot(ax=ax, color=color, linewidth=0.2, edgecolor='white')

# Plot LGAs with no data in gray
nodata = merged_ratio_map[merged_ratio_map['affordability'].isna()]
if len(nodata) > 0:
```

```
    nodata.plot(ax=ax, color="#dddddd", linewidth=0.2, edgecolor='white')

# Manual legend

patches = [mpatches.Patch(color=color, label=cat) for cat, color in colors.items()]
if len(nodata) > 0:
    patches.append(mpatches.Patch(color="#dddddd", label="No data"))

ax.legend(handles=patches, title="Affordability", loc='lower left')


# Final touches

ax.set_title("LGA Rental Affordability Classification - Australia (2021)", fontsize=16)
ax.set_axis_off()
plt.tight_layout()
plt.show()
```

V. Correlation analysis $between rent_to_income ratio and other dimensions$

In this section, we will analysis the relationship between rent_to_income ratio with 4 dimensions, including: population, population density, house stock and distance between LGA to capitals cities. After all, a heatmap is plotted to visualize the relationship among these all metrics.

These 4 dimensions are often mentioned, analysed in several articles and studies about rent stress. Data for the first 3 dimensions can be retreivved from 2021 abs data. Distance to nearest captical cities are calculated.

1. Population and density

We will use G01 dataset from 2021 census which give us population and areas $square km$ for all LGAs. Density: Population/Area in Square Km

python

```
density_df = pd.read_csv("/Volumes/workspace/project-trial/rental_private_market/ABS_2021_Census_G01_LGA_Pop
```

python

```
density_df.head()
```

	OBJECTID	Local Government Areas 2021 code	Local Government Areas 2021 name	Area in square kilometres	Uniform Resource Identifier	Total Persons: Males	Total Persons: Females	Total Persons: Persons	Age groups: 0-4 years: Males	Age groups: 0-4 years: Females	Age groups: 0-4 years: Persons	Ag group 5-1 year Male
0	1	10050	Albury	305.6386	https://linked.data.gov.au/dataset/asgsed3/LGA...	27087.0	29007.0	56093.0	1837.0	1732.0	3568.0	3576
1	2	10180	Armidale Regional	7809.4406	https://linked.data.gov.au/dataset/asgsed3/LGA...	14005.0	15115.0	29124.0	819.0	748.0	1571.0	1810
2	3	10250	Ballina	484.9692	https://linked.data.gov.au/dataset/asgsed3/LGA...	22271.0	24019.0	46296.0	1144.0	1061.0	2205.0	2800
3	4	10300	Balranald	21690.7493	https://linked.data.gov.au/dataset/asgsed3/LGA...	1116.0	1098.0	2208.0	58.0	77.0	135.0	167
4	5	10470	Bathurst Regional	3817.8645	https://linked.data.gov.au/dataset/asgsed3/LGA...	21857.0	21705.0	43567.0	1281.0	1201.0	2479.0	2920

5 rows × 116 columns

python

```
## Checking null values
density_df.isnull().sum()
```

```
OBJECTID                                      0
Local Government Areas 2021 code              0
Local Government Areas 2021 name              0
Area in square kilometres                    19
Uniform Resource Identifier                   0
                                             ..
Count of Persons: in other dwellings: Females 1
Count of Persons: in other dwellings: Persons 1
globalid                                      0
Shape__Area                                  19
Shape__Length                                19
Length: 116, dtype: int64
```

python

```
# Checking null values of Area (square km)
missing_area = density_df[density_df["Area in square kilometres"].isnull()]

missing_area[["Local Government Areas 2021 code",
              "Local Government Areas 2021 name"]]
```

	Local Government Areas 2021 code	Local Government Areas 2021 name
129	19499	No usual address NSW
130	19799	Migratory - Offshore - Shipping NSW
211	29499	No usual address $Vic.$
212	29799	Migratory - Offshore - Shipping $Vic.$
291	39499	No usual address Qld
292	39799	Migratory - Offshore - Shipping Qld
364	49499	No usual address SA
365	49799	Migratory - Offshore - Shipping SA
505	59499	No usual address WA
506	59799	Migratory - Offshore - Shipping WA
536	69499	No usual address $Tas.$
537	69799	Migratory - Offshore - Shipping $Tas.$
557	79499	No usual address NT
558	79799	Migratory - Offshore - Shipping NT
560	89499	No usual address ACT
561	89799	Migratory - Offshore - Shipping ACT
563	99499	No usual address OT
564	99799	Migratory - Offshore - Shipping OT
565	ZZZZZ	Outside Australia

python

```
## Checking null value of population

missing_totalperson = density_df[density_df["Count of Persons: in other dwellings: Persons"].isnull()]

missing_totalperson[["Local Government Areas 2021 code",
              "Local Government Areas 2021 name"]]
```

	Local Government Areas 2021 code	Local Government Areas 2021 name
565	ZZZZZ	Outside Australia

python

```
density_df["density"]= density_df["Total Persons: Persons"] / density_df["Area in square kilometres"]
```

python

```
density = density_df[["Local Government Areas 2021 code","Local Government Areas 2021 name" ,"Total Persons:
```

python

```
density
```

	Local Government Areas 2021 code	Local Government Areas 2021 name	Total Persons: Persons	density
0	10050	Albury	56093.0	183.527212
1	10180	Armidale Regional	29124.0	3.729332
2	10250	Ballina	46296.0	95.461732
3	10300	Balranald	2208.0	0.101795
4	10470	Bathurst Regional	43567.0	11.411353
...	...	...	...	...
561	89799	Migratory - Offshore - Shipping ACT	0.0	NaN
562	99399	Unincorp. Other Territories	2494.0	23.554834
563	99499	No usual address OT	11.0	NaN
564	99799	Migratory - Offshore - Shipping OT	0.0	NaN
565	ZZZZZ	Outside Australia	NaN	NaN

566 rows × 4 columns

python

```
## Checking NaN value of density
nan_density_lgas = density[
    density["density"].isna()
]
```

python

```
nan_density_lgas
```

	Local Government Areas 2021 code	Local Government Areas 2021 name	Total Persons: Persons	density
129	19499	No usual address NSW	11371.0	NaN
130	19799	Migratory - Offshore - Shipping NSW	0.0	NaN
211	29499	No usual address $Vic.$	9089.0	NaN
212	29799	Migratory - Offshore - Shipping $Vic.$	38.0	NaN
291	39499	No usual address Qld	14670.0	NaN
292	39799	Migratory - Offshore - Shipping Qld	183.0	NaN
364	49499	No usual address SA	3377.0	NaN
365	49799	Migratory - Offshore - Shipping SA	12.0	NaN
505	59499	No usual address WA	8055.0	NaN
506	59799	Migratory - Offshore - Shipping WA	514.0	NaN
536	69499	No usual address $Tas.$	1377.0	NaN
537	69799	Migratory - Offshore - Shipping $Tas.$	19.0	NaN
557	79499	No usual address NT	3500.0	NaN
558	79799	Migratory - Offshore - Shipping NT	99.0	NaN
560	89499	No usual address ACT	605.0	NaN
561	89799	Migratory - Offshore - Shipping ACT	0.0	NaN
563	99499	No usual address OT	11.0	NaN
564	99799	Migratory - Offshore - Shipping OT	0.0	NaN
565	ZZZZZ	Outside Australia	NaN	NaN

affordability df is a shorterned dataframe from full merged data, including only 4 relevant fields.

python

```
affordability = merged[['Local Government Areas 2021 code','Local Government Areas 2021 name_x','rent_to_inc
```

python

```
## Merge affordabilty df with density df using Inner Join and LGA code as key word

merged = pd.merge(affordability, density, on="Local Government Areas 2021 code", how="inner")
```

The merged data now has 3 dimensions that we need, rent_to_income ratio, populationa and density, preparing for correlation analysis

python

```
#Merged data now include 3 important dimension, rent_to_income ratio, population and density
merged
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	rent_to_income_ratio	affordability	Local Government Areas 2021 name	Total Persons: Persons	density
0	10050	Albury	0.328682	Moderately Unaffordable	Albury	56093.0	183.527212
1	10180	Armidale Regional	0.347865	Moderately Unaffordable	Armidale Regional	29124.0	3.729332
2	10250	Ballina	0.495655	Severely Unaffordable	Ballina	46296.0	95.461732
3	10300	Balranald	0.244194	Affordable	Balranald	2208.0	0.101795
4	10470	Bathurst Regional	0.355311	Moderately Unaffordable	Bathurst Regional	43567.0	11.411353
...	...	...	...	...	...	...	...
551	79799	Migratory - Offshore - Shipping NT	NaN	No data	Migratory - Offshore - Shipping NT	99.0	NaN
552	89399	Unincorporated ACT	0.375219	Moderately Unaffordable	Unincorporated ACT	453890.0	192.478541
553	89799	Migratory - Offshore - Shipping ACT	NaN	No data	Migratory - Offshore - Shipping ACT	0.0	NaN
554	99399	Unincorp. Other Territories	0.304816	Moderately Unaffordable	Unincorp. Other Territories	2494.0	23.554834
555	99799	Migratory - Offshore - Shipping OT	NaN	No data	Migratory - Offshore - Shipping OT	0.0	NaN

556 rows × 7 columns

2. House stock $totalhousehold$

House stock is derived from G33 data set from 2021 ABS census

python

```
total_household_df= pd.read_csv("/Volumes/workspace/project-trial/rental_private_market/ABS_2021_Census_G33_
```

python

```
total_household_df
```

	OBJECTID	Local Government Areas 2021 code	Local Government Areas 2021 name	Area in square kilometres	Uniform Resource Identifier	Household Income: Negative Nil income: Family households	Household Income: Negative Nil income: Non family households	Household Income: Negative Nil income: Total	Household Income: $1-149$: Family households	Household Income: $1-149$: Non family households
0	1	10050	Albury	305.6386	https://linked.data.gov.au/dataset/asgsed3/LGA...	98.0	200.0	300.0	43.0	74.0
1	2	10180	Armidale Regional	7809.4406	https://linked.data.gov.au/dataset/asgsed3/LGA...	72.0	90.0	161.0	32.0	52.0
2	3	10250	Ballina	484.9692	https://linked.data.gov.au/dataset/asgsed3/LGA...	92.0	162.0	252.0	59.0	84.0
3	4	10300	Balranald	21690.7493	https://linked.data.gov.au/dataset/asgsed3/LGA...	6.0	6.0	16.0	0.0	0.0
4	5	10470	Bathurst Regional	3817.8645	https://linked.data.gov.au/dataset/asgsed3/LGA...	80.0	148.0	231.0	38.0	63.0
...	...	...	...	...	...	...	...	...	...	...
561	562	89799	Migratory - Offshore - Shipping ACT	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0
562	563	99399	Unincorp. Other Territories	105.8806	https://linked.data.gov.au/dataset/asgsed3/LGA...	7.0	15.0	15.0	0.0	10.0
563	564	99499	No usual address OT	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0
564	565	99799	Migratory - Offshore - Shipping OT	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	0.0	0.0	0.0	0.0	0.0
565	566	ZZZZZ	Outside Australia	NaN	https://linked.data.gov.au/dataset/asgsed3/LGA...	NaN	NaN	NaN	NaN	NaN

566 rows × 68 columns

python

```
#Checking null values of house stock
total_household_df.isnull().sum()
```

```
OBJECTID                                          0
Local Government Areas 2021 code                  0
Local Government Areas 2021 name                  0
Area in square kilometres                        19
Uniform Resource Identifier                       0
                                                 ..
Household Income: Total: Non family households    1
Household Income: Total: Total                    1
globalid                                          0
Shape__Area                                      19
Shape__Length                                    19
Length: 68, dtype: int64
```

python

```
merged
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	rent_to_income_ratio	affordability	Local Government Areas 2021 name	Total Persons: Persons	density
0	10050	Albury	0.328682	Moderately Unaffordable	Albury	56093.0	183.527212
1	10180	Armidale Regional	0.347865	Moderately Unaffordable	Armidale Regional	29124.0	3.729332
2	10250	Ballina	0.495655	Severely Unaffordable	Ballina	46296.0	95.461732
3	10300	Balranald	0.244194	Affordable	Balranald	2208.0	0.101795
4	10470	Bathurst Regional	0.355311	Moderately Unaffordable	Bathurst Regional	43567.0	11.411353
...	...	...	...	...	...	...	...
551	79799	Migratory - Offshore - Shipping NT	NaN	No data	Migratory - Offshore - Shipping NT	99.0	NaN
552	89399	Unincorporated ACT	0.375219	Moderately Unaffordable	Unincorporated ACT	453890.0	192.478541
553	89799	Migratory - Offshore - Shipping ACT	NaN	No data	Migratory - Offshore - Shipping ACT	0.0	NaN
554	99399	Unincorp. Other Territories	0.304816	Moderately Unaffordable	Unincorp. Other Territories	2494.0	23.554834
555	99799	Migratory - Offshore - Shipping OT	NaN	No data	Migratory - Offshore - Shipping OT	0.0	NaN

556 rows × 7 columns

total_households is short data frame of original total_household_df with only 2 columns, LGA code and total household

python

```
total_households = total_household_df[["Local Government Areas 2021 code", "Household Income: Total: Total"]
```

Merge total_households with merged data $already has 3 dimension, rent_to_income ratio, population, density$

python

```
merged = pd.merge(merged,total_households, on="Local Government Areas 2021 code", how="inner")
```

Merged data now is extended and have already has 4 dimension, rent_to_income ratio, population, density and total household, preparing for correlation analysis

python

```
merged
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x	rent_to_income_ratio	affordability	Local Government Areas 2021 name	Total Persons: Persons	density	Household Income: Total: Total
0	10050	Albury	0.328682	Moderately Unaffordable	Albury	56093.0	183.527212	22182.0
1	10180	Armidale Regional	0.347865	Moderately Unaffordable	Armidale Regional	29124.0	3.729332	10636.0
2	10250	Ballina	0.495655	Severely Unaffordable	Ballina	46296.0	95.461732	18281.0
3	10300	Balranald	0.244194	Affordable	Balranald	2208.0	0.101795	733.0
4	10470	Bathurst Regional	0.355311	Moderately Unaffordable	Bathurst Regional	43567.0	11.411353	15942.0
...	...	...	...	...	...	...	...	...
551	79799	Migratory - Offshore - Shipping NT	NaN	No data	Migratory - Offshore - Shipping NT	99.0	NaN	3.0
552	89399	Unincorporated ACT	0.375219	Moderately Unaffordable	Unincorporated ACT	453890.0	192.478541	168400.0
553	89799	Migratory - Offshore - Shipping ACT	NaN	No data	Migratory - Offshore - Shipping ACT	0.0	NaN	0.0
554	99399	Unincorp. Other Territories	0.304816	Moderately Unaffordable	Unincorp. Other Territories	2494.0	23.554834	986.0
555	99799	Migratory - Offshore - Shipping OT	NaN	No data	Migratory - Offshore - Shipping OT	0.0	NaN	0.0

556 rows × 8 columns

python

```
## Rename population and total house hold

merged = merged.rename(columns={
    "Total Persons: Persons": "total_population",
    "Household Income: Total: Total": "total_houses"
})
```

3. Distance from LGAs to capital cities

This data is calculated, final merged data will have additioned distance to capital cities, which are the crow-flies from central point of LGA to nearest captical city

python

```python
import geopandas as gpd
from shapely.geometry import Point

# Create a GeoDataFrame of capital cities (coordinates in GDA2020)

# Coordinates in (longitude, latitude)
capital_coords = {
    "Sydney": (151.2093, -33.8688),
    "Melbourne": (144.9631, -37.8136),
    "Brisbane": (153.0251, -27.4698),
    "Perth": (115.8605, -31.9505),
    "Adelaide": (138.6007, -34.9285),
    "Hobart": (147.3272, -42.8821),
    "Darwin": (130.8456, -12.4634),
    "Canberra": (149.1300, -35.2809)
}

# Convert dictionary to GeoDataFrame
capital_points = gpd.GeoDataFrame(
    {"city": list(capital_coords.keys())},
    geometry=[Point(xy) for xy in capital_coords.values()],
    crs="EPSG:4326"
)

# Ensure both datasets use the same projected CRS for distance
# (use Australian Albers projection for accurate metre distances)

merged_ratio_map = merged_ratio_map.to_crs("EPSG:3577")   # Australia Albers
capital_points = capital_points.to_crs("EPSG:3577")

# Compute centroid of each LGA

merged_ratio_map["centroid"] = merged_ratio_map.geometry.centroid

# Compute distance from each LGA centroid to ALL capital cities

distances = merged_ratio_map["centroid"].apply(
    lambda c: capital_points.distance(c).min()      # min = nearest capital
)

# convert metres → km
merged_ratio_map["dist_to_capital_km"] = distances / 1000


print(merged_ratio_map[["Local Government Areas 2021 code", "Local Government Areas 2021 name_x",
                        "dist_to_capital_km"]].head())
```

```
  Local Government Areas 2021 code  ... dist_to_capital_km
0                            10050  ...         212.180068
1                            10180  ...         342.590078
2                            10250  ...         161.474560
3                            10300  ...         445.828453
4                            10470  ...         162.778257

[5 rows x 3 columns]
```

python

```
distance_to_capital=merged_ratio_map[["Local Government Areas 2021 code", "Local Government Areas 2021 name_
```

Merging distance_to_capital and previous merged dataset will give us extented dataset, with the 5th $final$ dimension added, which is distance to captial city

During merging process, below error occured as there were already 2 datasets with column "Local Government Areas 2021 name_x_" and python can not add another "_x" into the column name.

"merged = pd.merge$merged, distance_to_capital, on = "Local Government Areas 2021 code", how = "inner"$
MergeError: Passing 'suffixes' which cause duplicate columns {'Local Government Areas 2021 name_x_x'} is not allowed."

Hence, we need use suffixes=$" - left", "right"$ to LGA name column

python

```
merged = pd.merge(merged,distance_to_capital, on="Local Government Areas 2021 code", how="inner", suffixes=(
```

python

```
merged.head()
```

	Local Government Areas 2021 code	Local Government Areas 2021 name_x-left	rent_to_income_ratio	affordability	Local Government Areas 2021 name	total_population	density	total_houses	Local Government Areas 2021 name_x_right	dist_to_capital_km
0	10050	Albury	0.328682	Moderately Unaffordable	Albury	56093.0	183.527212	22182.0	Albury	212.180068
1	10180	Armidale Regional	0.347865	Moderately Unaffordable	Armidale Regional	29124.0	3.729332	10636.0	Armidale Regional	342.590078
2	10250	Ballina	0.495655	Severely Unaffordable	Ballina	46296.0	95.461732	18281.0	Ballina	161.474560
3	10300	Balranald	0.244194	Affordable	Balranald	2208.0	0.101795	733.0	Balranald	445.828453
4	10470	Bathurst Regional	0.355311	Moderately Unaffordable	Bathurst Regional	43567.0	11.411353	15942.0	Bathurst Regional	162.778257

python

```python
##Correlation analysis

cols = [
    "rent_to_income_ratio",
    "density",
    "total_population",
    "total_houses",
    "dist_to_capital_km",
]

correlation = merged[cols].corr()
print(correlation)
```

```
                     rent_to_income_ratio  ...  dist_to_capital_km
rent_to_income_ratio              1.000000  ...           -0.416392
density                           0.186589  ...           -0.303679
total_population                  0.395964  ...           -0.265397
total_houses                      0.383273  ...           -0.270341
dist_to_capital_km               -0.416392  ...            1.000000

[5 rows x 5 columns]
```

python

```python
plt.figure(figsize=(8,5))
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    vmin=-1,
    vmax=1
)
plt.title("Correlation Matrix")
plt.show()
```

Result of correlation and heatmap analysis $Relationship from strongest to weakest$

- Distance to capital and rent_to_income ratio: Coefficient **−0.42**, means moderate negative relationship. LGAs closer to capital cities tend to have higher rent stress. As distance increases, rent-to-income ratios fall.
- Population and rent_to_income ratio: Coefficient **0.4**, means moderately higher rent stress in more populous LGAs.
- Total house and rent_to_income ratio: Coefficient **0.38**, means similar moderate effect; larger housing markets face more rent pressure.
- Density and rent_to_income ratio: Coeficient: **0.19**, means weak relationship; density alone is not a strong driver of rent stress.

High density does not automatically mean high rent stress. Distance to economic centres matters more

Regression between rent_to_income ratio to distance to capital cities and population $two strongest relationships$

python

```python
sns.set(style="whitegrid", font_scale=1.2)

# Keep only rows with valid values
plot_df = merged_ratio_map.dropna(subset=["rent_to_income_ratio", "dist_to_capital_km"])

plt.figure(figsize=(10,6))
# Scatter points
sns.scatterplot(
    data=plot_df,
    x="dist_to_capital_km",
    y="rent_to_income_ratio",
    hue="affordability",
    palette={
        "Affordable": "#2ca02c",
        "Moderately Unaffordable": "#ff7f0e",
        "Severely Unaffordable": "#d62728",
        "No data": "#dddddd"
    },
    alpha=0.7
)
# Regression line 1
sns.regplot(
    data=plot_df,
    x="dist_to_capital_km",
    y="rent_to_income_ratio",
    scatter=False,
    color="blue",
    line_kws={"linestyle":"--"}
)
plt.xlabel("Distance to Nearest Capital City (km)")
plt.ylabel("Rent-to-Income Ratio")
plt.title("Rent-to-Income Ratio vs Distance to Capital City - Australia (2021)")
plt.legend(title="Affordability", loc="upper right")
plt.tight_layout()
plt.show()
```

python

```python
sns.set(style="whitegrid", font_scale=1.2)

# Keep only rows with valid values
plot_df = merged.dropna(subset=["rent_to_income_ratio", "total_population"])
plt.figure(figsize=(10,6))
sns.regplot(
    data=plot_df,
    x="total_population",
    y="rent_to_income_ratio",
    scatter=False,             # only line
    color="blue",
    line_kws={"linestyle":"--", "linewidth":2}
)
# Scatter points colored by affordability
sns.scatterplot(
    data=plot_df,
    x="total_population",
    y="rent_to_income_ratio",
    hue="affordability",
    palette={
        "Affordable": "#2ca02c",
        "Moderately Unaffordable": "#ff7f0e",
        "Severely Unaffordable": "#d62728",
        "No data": "#dddddd"
    },
    alpha=0.7
)
# Customize plot
plt.xlabel("Total Population")
plt.ylabel("Rent-to-Income Ratio")
plt.title("Rent-to-Income Ratio vs Total Population - Australia (2021)")
plt.legend(title="Affordability", loc="upper right")
plt.tight_layout()

# Show plot
plt.show()
```

VI. Conclusion and Recommendation

1. Conclusion

The findings highlight a structural spatial dimension to rental affordability in Australia. LGAs closer to capital cities face heightened rent stress due to the concentration of employment, services, and infrastructure, which intensifies housing demand. While denser areas tend to be closer to capital cities, density itself does not emerge as a dominant factor once spatial proximity is considered.

If distance dominates: - Rent stress is a spatial access problem.

If density dominates: Rent stress is a supply/urban form problem

The project result conludes **distance** dominates over density in playing a role as driver for rent affordability.

The result is in line with other studies in Australia.

- **AHURI Final Report No. 353 — Urban productivity and affordable rental housing supply in Australian cities and regions**. This major AHURI study directly examines the relationship between affordable rental housing supply and proximity to employment and labour markets in Australia's capital cities. The research finds that lower-income rental households in Sydney and Melbourne sacrifice housing affordability to remain physically close to employment opportunities, indicating that distance from job centres is a key factor in rental stress patterns rather than density alone. It also shows that renter workers who live farther out, in satellite cities, experience higher commuting burdens to access jobs — highlighting the trade-off between affordability and location/access.

SGS Economics & Planning / National Shelter Rental Affordability Index RAI

While not an academic journal article, the RAI is one of Australia's most authoritative ongoing analyses of rental affordability, used by governments and academics. Its findings are relevant to location-linked affordability patterns: Capital cities $especiallySydney, Perth, Melbourne$ consistently register the lowest RAI scores, meaning rents relative to incomes are most stretched in the geographic cores of employment and services. Regional areas sometimes score better or worse, depending on how wage structures and demand from migrants $oftentornbetweencapitalcityincomeadvantagesandregionallivingcosts$ interact — which highlights that distance and access to economic opportunities matter for affordability, more so than raw density metrics.

** 2. Policty Implication**

a) Reduce demand pressure in capital cities
Given the strong association between rent stress and proximity to capital cities, policies that decentralise employment and services are likely to be effective. This includes:
• investing in regional employment hubs,
• relocating public sector agencies to regional centres, and
• incentivising private firms to operate outside metropolitan cores.
By reducing the necessity for households to locate near capital cities for employment, these measures can alleviate demand pressures in high-stress rental markets.

b) Improve transport connectivity
Enhanced transport infrastructure linking outer-metropolitan and regional LGAs to capital cities can reduce effective distance without intensifying inner-city housing demand. Faster, more reliable transport allows households to access metropolitan labour markets while residing in lower-cost areas, mitigating rent-to-income pressures.

c) Shift focus from density targets to affordability outcomes
The weak correlation between density and rent stress suggests that density targets alone are insufficient as an affordability strategy. Planning frameworks should prioritise:
• the provision of affordable rental dwellings,
• tenure diversity, and
• inclusionary zoning mechanisms, rather than density increases in isolation.

d) Expand affordable and social housing near employment centres
While decentralisation is important, demand for capital-city locations will persist. Increasing the supply of non-market and affordable rental housing in well-located LGAs can directly reduce rent stress among low- and middle-income households without relying solely on market supply responses.

e) Targeted rent assistance and income support
Given that rent stress reflects both housing costs and income constraints, policies such as:

• indexed Commonwealth Rent Assistance,
• targeted subsidies in high-stress LGAs, can provide short-term relief while longer-term structural reforms take effect.

3. Key takeaways

This study demonstrates that rent stress across Australian LGAs is more strongly associated with proximity to capital cities than with population density. The results challenge density-focused affordability narratives and underscore the importance of spatial economic structure in shaping rental outcomes. Effective policy responses must therefore combine decentralisation, transport investment, and targeted affordable housing provision to address the root causes of rental stress.

VII. LIMITATIONS

The project use ABS 2021 census data, which is quite outdated. Plus, the data was already been cleaned, aggregated, which didn't give project executer chancce to show case data cleaning and manipulation process.

VIII FURTURE DEVELOPMENT

Combined with 2026 abs data to analyze the trends and changes.